

Scale Fixes Local Errors First: Differential Scaling of Chess Transformer Failures

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Abstract

Scaling reduces how often language models err over long horizons, but it is rarely asked which errors it removes. We classify every illegal move made by search-free chess transformers by what made it illegal, using only a rules engine, and measure how each class’s absolute rate falls with model size. Across 18 models spanning 3.4M to 39.9M parameters, errors decidable from the move’s own squares fall with exponent -0.570 , while moves illegal only because they leave the king in check fall with exponent -0.114 : a paired differential of $+0.456$ [$+0.250$, $+0.604$]. Every class improves; the rates differ. Failures occurring despite correctly decoded local state have a nearly flat joint rate, because improving decoding offsets a conditional failure rate falling with exponent -0.125 . The ordering replicates on an independent character-level ladder spanning a 38-fold parameter range, and in OthelloGPT 160 of 161 residual failures are of the non-local kind. Internal belief is coupled to action locally, and the repair we test installs about two squares but no reliable whole-board correction. Two reasonable scores of the same intervention then disagree: illegal-move rate ranks a norm-matched random edit above correct repair in 18 of 18 models, while the repair shifts a fixed forced choice further, so one headline number cannot settle whether an intervention helped. Finally, these per-category fits and a separately fitted aggregate cannot both hold, since a sum of power laws with differing exponents is generally not a power law; we derive the total from the categories instead.

Keywords: scaling, failure analysis, world models, state tracking, chess, interpretability, evaluation methodology

1 Introduction

Scaling laws describe how aggregate loss falls with parameters and data (Kaplan et al., 2020; Hoffmann et al., 2022). They say much less about the composition of what remains. A model that makes half as many errors at ten times the size may have removed every kind of error evenly, or removed one kind almost entirely and left another untouched. The two cases recommend different research. If scale removes errors evenly, more scale is the remedy for all of them. If it removes some and not others, the ones it leaves are where effort should go, and interventions aimed at the kind scale already removes are buying something that arrives for free.

Long-horizon reasoning is where this question matters most and is hardest to ask, because in open-ended text there is no ground truth for why a step was wrong. We ask it where there is. A search-free transformer trained on chess move notation produces a move at every position, a rules engine decides whether it is legal, and when it is not, the same engine says why. Some illegal moves are decidable from the two squares the move names: the source square is empty, or holds an opponent’s piece, or the destination holds one of the mover’s own. Others turn on the whole board: the move is geometrically possible for the piece but its path is blocked, or it is pseudo-legal

and rejected only because it leaves the king in check. That distinction between local and global illegality is exact, needs no model internals, and is the axis along which we measure scaling.

What we find.

1. **Failure classes scale at different rates, ordered by locality.** Absolute rates of local errors fall steeply with size and global ones slowly. The empty-source class falls 4.01-fold across our ladder and the check class 1.31-fold, a differential exponent of +0.461 that excludes zero in every rung resample. No probe, intervention, or interpretability method enters this result.
2. **It is not an artefact of our models.** The same ordering holds on an independent ladder with different data, notation, and training code, and in a game whose legality is non-local by construction.
3. **The slowest-improving component is the one local correction cannot reach.** Every disjoint failure class improves with scale; they differ in how fast. Internal belief is causally coupled to action at the squares a move touches, the repair procedure we test installs about two squares reliably and whole-board correction not at all, and failures occurring despite correct belief at those squares have a nearly flat joint rate because improving decoding offsets a falling conditional failure rate.
4. **The usual way of scoring interventions reads this backwards.** Illegal-move rate rewards edits that flatten the output distribution, and on our data it prefers a random direction to the correct repair in every model.

Contributions. A probe-free, rules-grounded measurement of how failure composition changes with scale; its replication across two independent ladders, with a consistency check in a second game; a mechanistic account, with a measured capability limit on the causal instrument; and a demonstration that a widely used intervention metric is anti-correlated with the effect it is meant to measure. The probe and edit instruments used in Section 7 are established in prior work (Rathore et al., 2026); here they serve only to explain a result that stands without them.

2 Related Work

Scaling and what it leaves. Aggregate loss follows power laws in parameters, data, and compute (Kaplan et al., 2020; Hoffmann et al., 2022). Decomposing an aggregate into components with separate exponents is the step we take, applied to error classes rather than to loss, in a domain where the classes are defined by an oracle rather than by annotation.

World models in game transformers. Linear probes recover board state from move sequences (Li et al., 2023), with linear structure (Nanda et al., 2023); chess is an established state-tracking testbed (Toshniwal et al., 2022); and board-state probes in a character-level chess model can be intervened on (Karvonen, 2024), later serving as ground truth for dictionary learning (Karvonen et al., 2024). Walker and Lyons (2026) benchmark exact state tracking across architectures at a parameter range close to ours, Li and Jiang (2026) separate state tracking from decision quality, and Pereira and Zuidema (2026) and Harang et al. (2025) document state decay over long generations. State decay and chess state tracking are not our contribution; how the composition of failure changes with size is.

Coupling between probes and behaviour. Balogh and Jelasity (2026) find the fraction of illegal moves legal on a probe-reconstructed board small in large-data models, and a probe gradient

nearly orthogonal to the output head. Section 7 reconciles this with a strong causal edit effect: coupling is local, so whole-board measures understate it and square-level ones do not.

Evaluating interventions. Inference-time search and verification (Yao et al., 2023; Lightman et al., 2024; Wang et al., 2023) and causal editing (Meng et al., 2022) are commonly scored by an end-task error rate. Section 8 shows why that rate can reward perturbations for the wrong reason.

3 Setup

Models. Our ladder has 6 decoder-only transformers from 3.4M to 39.9M parameters, each trained with 3 seeds, for 18 runs. Models read one token per move in UCI notation and are trained on human games, with evaluation on held-out games and no search at inference. Depth and width vary together along the ladder, so a size exponent here is not a pure depth or width exponent (Section 11).

Training. Every figure in this paragraph is read out of the checkpoints rather than transcribed, by the same macro pipeline as the results, so a hyperparameter cannot quietly stop describing the run it labels.

Rung	Layers	Width	Heads	Params (M)	Val. loss
6L192	6	192	6	3.4	3.342
8L256	8	256	8	7.3	3.029
12L256	12	256	8	10.5	2.917
8L384	8	384	8	15.7	2.840
12L384	12	384	8	22.8	2.754
12L512	12	512	8	39.9	2.709

All 18 runs share one recipe: AdamW at learning rate 3×10^{-4} , weight decay 0.01, $\beta = (0.9, 0.95)$, a one-cycle schedule with 5 percent warmup, gradient norm clipped to 1, and mixed precision with loss scaling. Each run takes 12,000 updates at batch size 32, which is 30.0 million supervised tokens over a corpus of 96,000 games of mean length 78 tokens, or about 4.0 epochs. Validation loss is evaluated every 500 updates on held-out games.

Two consequences deserve stating because they bound what the exponents mean. The token budget is held fixed across the ladder, so larger models are not given proportionally more data: these are parameter exponents at constant data, not compute-optimal scaling exponents, and they are not comparable to exponents fitted along a compute-optimal frontier. And the reported checkpoint is the final one, with no early stopping and no best-validation selection. That choice is innocuous here rather than by assumption: the final checkpoint is also the lowest-validation-loss checkpoint in 18 of 18 runs, so selection cannot be carrying any part of the result. Validation loss falls monotonically along the ladder, which confirms the ladder is one of capability and not merely of parameter count. The table lists the seed-zero run of each rung; over all 18 runs the loss spans 2.700 to 3.349, and the two sets of numbers differ only because one is a single run and the other the extremes across seeds.

Failure taxonomy. At every position from ply 20 onward we take the model’s top-1 move and ask the rules engine whether it is legal. Each illegal move is assigned to exactly one class, in priority order, by what made it illegal: *empty source* (no piece on the source square), *opponent*

source (the source holds the other side’s piece), *own destination* (the destination holds the mover’s own piece), *geometry* (a piece of the right colour is present but cannot reach the destination, or its path is blocked), and *leaves check* (pseudo-legal, rejected only because the king would be in check). The first three are decidable from the two named squares. The ordering of the remaining two is measured rather than asserted, below, because an exponent is a measurement and relating it to an ordering we merely argued for would be the weaker half of the comparison.

How much board each class actually depends on. For a position and an illegal move, call a square a *dependency* if altering its contents alone can change whether the move is legal, and let k count such squares, excluding the two the move names because every class depends on those by construction. The definition is causal: it asks what the verdict turns on, not what the rule text mentions. It is computed by brute force over all 64 squares and all 12 piece types, on positions from real games with no model involved, so k is a property of the rules and the position rather than of any network. The edited boards are not constrained to be reachable or even valid: python-chess answers legality queries on such positions as what its documentation calls pseudo-chess, and a board may end up without a king or with the side not to move in check. We allow this deliberately, because the question is which squares the legality rule reads, not which positions a game could reach. It does mean k counts sensitivity to physically unreachable alterations, and the castling defect above was found precisely by asking why a class decidable from one square appeared to depend on many. Over 120 games and 4,204 sampled illegal moves drawn from the five classes of the original partition, plus 603 castling attempts sampled separately once the audit below identified them as a distinct class, the dependency counts are as follows. The two geometry rows subdivide the geometry row above them rather than adding to it, so the partition total is the five original classes alone.

Class	n	mean k	max k
empty source	960	0.00	0
opponent source	960	0.00	0
own destination	949	0.00	0
geometry, impossible	932	0.00	0
geometry, blocked path	28	0.64	1
castling	603	0.13	7
leaves check	375	17.49	55

Two things follow, and the second is inconvenient for us. The local and global distinction this paper rests on is confirmed, and is now a measurement rather than an argument: the check class depends on 17.49 squares on average, with a standard deviation of 11.3 and a maximum of 55, while every other class decided by the move’s own squares depends on at most one. Castling sits in between, with a maximum of 7: rights and the intervening squares can each be interfered with, which is part of why separating it from the destination-occupancy class matters. The separation between the locally and globally decided classes is not a matter of degree.

But k does *not* explain the ordering of the exponents. Geometry has an exponent of -0.207 , roughly halfway between the local classes and the check class, and a dependency set indistinguishable from the local ones. The reason lies in the taxonomy rather than in the models, and the table above splits it: of 960 geometry errors sampled, 97.1 percent are moves the piece cannot make under any arrangement of the other squares, with k exactly zero, and only 2.9 percent are genuinely blocked paths. One label covered two mechanically different errors, and the usual description of geometry as depending on intervening squares holds for the smaller part of the cases carrying it.

An audit of the own-destination maximum, and what it found. An earlier version of this table reported a maximum of 48 for the own-destination class, which should be decidable from the destination square alone and so should have a maximum of zero. The discrepancy was real and it was ours. King onto one’s own rook is how both UCI and Chess960 encode castling, so a destination-occupancy test collects castling attempts; and because `python-chess` enumerates castling as `extttelg1` while also accepting the `extttelh1` form, a membership test against the legal moves let a genuinely *legal* castling move into a sample of illegal ones. Nearly any edit to such a board removes castling rights, blocks the path, or places the king in check, which is where 48 came from. Castling is now its own class, legality is tested directly rather than by membership, and the own-destination maximum is 0 over 949 moves, as the definition requires. We report the defect because the same classifier labels the model-generated failures in Table 1: an illegal castling attempt there is counted as an own-destination error. The individual moves were not retained, so the affected share cannot be recovered without regenerating the failure sets, and we flag it as a known mislabel rather than correct it silently.

What k measures, and what it does not. The statistic counts squares whose alteration *alone* flips the verdict, which is single-square sufficiency and not the information the rule reads. The gap is not hypothetical and it has a predictable shape: a path held by two pieces flips on neither one alone, so it scores zero while plainly depending on both. That is a testable prediction about which moves score zero, and it holds exactly. Among the 28 blocked-path moves, those with a single blocker have $k = 1$ in 18 of 19 cases, and every one of the 9 moves with two or more blockers has $k = 0$. So k is a lower bound on dependence, its deficit is characterised rather than unknown, and in this class the deficit is fully accounted for by blocker count. This does not automatically strengthen the comparison between classes, because the amount of undercounting can differ between them: a class with many multi-piece dependencies loses more than one with few. What it does establish is a lower bound on each class separately, and the bound is already decisive for the one that matters, since the check class reaches 17.49 under a measure that systematically undercounts. We report this rather than choose k to fit. Hand-assigned values of k produce a coefficient of determination anywhere from about two thirds to about seven eighths depending on which values are assigned, which measures the freedom in the assignment and not a relationship in the data. The ordinal claim is supported at the extremes, where it is now measured. A continuous law predicting an exponent from locality is not supported, and we assert none. Across all 18 runs the residual class that none of these rules catch has a maximum share of 0.0000, so the taxonomy is exhaustive over the failures observed. We previously wrote that this also confirmed the classifier correct. It does not: assigning every case to some class says nothing about whether the assignments are right, and the defect described in Section 4 was present throughout while the residual class stayed empty.

Absolute rates, not shares. We report each class as a fraction of all scored positions, its share of illegal moves multiplied by the illegal-move rate. Shares are misleading here: the illegal-move rate itself falls with size, so a class whose absolute rate falls can still grow as a share of what remains. We state this because an earlier draft of this analysis read a rising share as the model getting worse at a class it was in fact getting better at.

Exponents and intervals. A class’s exponent is the slope of log absolute rate on log parameter count, fitted to all runs. Intervals resample the 6 rungs with replacement, since seeds within a rung are not independent draws of the architecture. A differential between two classes is bootstrapped on paired resamples, so its interval is on the difference itself.

A resampling interval over 6 rungs does not show whether one rung carries the result, so we also refit with each rung deleted in turn. The headline differential between the check class and empty-source errors is +0.461 on the full ladder and stays within [+0.360, +0.539] under every single-rung deletion, the largest shift being 0.101. The sign is the same in all 6 refits. Both the smallest and the largest rung can be removed without changing the conclusion, which is the specific worry a short ladder invites.

Computing environment. Every run in this paper, training and analysis alike, fits on a single NVIDIA GeForce RTX 3050 Laptop GPU with 4 GiB of device memory, in a host with 15 GB of physical memory visible to the operating system, under Windows 11 (build 26200), using Python 3.12.10 with PyTorch 2.5.1+cu121 on CUDA 12.1, NumPy 2.4.6, SciPy 1.18.1, and python-chess 1.11.2 as the rules engine. That memory ceiling, not a choice about what was interesting to measure, is what sets the top rung of the ladder at 39.9M parameters, and it is why the external ladders of Section 6 carry the weight they do: they extend the size range past what we can train ourselves. Versions are recorded by `code/record_environment.py` rather than typed here, so they cannot go stale if the work moves to another machine.

4 The Corrected Classification Pass

The audit above found two defects in the classifier, and the second is more serious than a mislabel. Disclosing them is not the same as correcting them, so the failure sets were regenerated from the saved checkpoints and every downstream quantity recomputed. This section reports what changed, what did not, and how we know which is which.

The two defects. First, king onto one’s own rook is how both UCI and Chess960 encode castling, so a test for a piece of one’s own colour on the destination square collects castling attempts and files them as destination-occupancy errors. Second, and worse, the test deciding whether a move counted as a failure at all asked whether the move was absent from `board.legal_moves`, the generator python-chess uses to enumerate legal moves. That generator emits castling as `e1g1`, while `board.is_legal` also accepts the `e1h1` king-onto-rook form of the same move. A legal castling move written the second way is therefore absent from the enumeration while being legal, and would be recorded as a failure. We state the distinction precisely because it is the whole mechanism: the original gate tested membership in the materialised set of generated moves, not legality, and the two differ exactly on this encoding.

The corrected pass. The gate now calls `is_legal` directly, castling is its own class, geometry is split into impossible movement and blocked path, and every failure is recorded with its position, its move and its label under both the old and the new scheme. That last point is why the affected share previously could not be quantified: the individual moves were not retained and only aggregate shares were stored. They are retained now.

Labelling each move both ways on the same recorded position matters for attribution. The difference between the two labels is then caused by the classifier alone, and cannot be confounded with any numerical difference between this pass and the original.

No legal move was ever counted as a failure. Across 16,200 recorded failures from all 18 runs, 0 were legal under the corrected gate. The failure mode that would have been most damaging, legal moves inflating the failure counts and with them the aggregate rates, did not occur. The reason is

visible once the mechanism is stated: for the defect to bite, a model must emit the king-onto-rook form while castling is actually available, and these models are trained on move tokens produced by the same library’s standard encoding, so they learn `e1g1` and not `e1h1` as the way to castle. That token is in the vocabulary only as an ordinary rook move. Aggregate illegal-move rates, and every quantity built on them, are therefore unaffected.

What was relabelled, and what the headline rests on. 5,567 of the 16,200 moves changed class: 4,106 and 1,352 moved from the geometry label into its two sub-classes, which is a refinement rather than a correction, and 109 genuinely illegal castling attempts left the destination-occupancy class, which is the correction. No move entered or left either of the two classes the headline differential is computed from: 0 for empty-source errors and 0 for check violations. The differential is therefore unchanged by the relabelling as a matter of arithmetic rather than of estimation, which is the specific thing a reader needs in order to know that the central result survives the correction.

The cost of regenerating on different hardware. The original pass ran in float16 on a GPU that is no longer attached; the corrected pass ran in float32 on CPU. Comparing the *old* labels from the new pass against the stored shares isolates that change, since the classifier is then held fixed. Class shares move by 0.0011 on average and at most 0.0121, which is what changing precision does to the handful of positions where the top two moves are nearly tied. We report this rather than assume a rerun reproduces the original. The headline differential is +0.456 [+0.250, +0.604] in the corrected pass against +0.461 as published, and that residual difference is this drift rather than the classifier.

The repaired taxonomy changes what geometry means. Splitting geometry is not cosmetic. The two sub-classes scale differently: impossible movement has exponent -0.254 [-0.313, -0.220], while blocked paths have -0.053 [-0.228, +0.126], an interval that includes zero. The aggregate geometry exponent was averaging the two, which is the same objection this paper raises against fitting one exponent to a total whose parts scale differently, turned on one of our own categories. Against the locally decided classes the check differential narrows but survives: check against impossible movement is +0.139 [+0.069, +0.234]. The blocked-path exponent is poorly determined on 6 rungs and we draw no ordering conclusion from it.

5 Differential Scaling

Every class falls with size, so the result is not that larger models get worse at anything. It is that they get better at different things at very different rates (Table 1, Figure 1a). The exponents order themselves by how local the rule is. Empty-source errors fall with exponent -0.574 ; geometry, which depends on the squares between source and destination, with -0.207 ; and check violations, which depend on every attacker’s position, with -0.112 . The paired differential between the check and empty-source exponents is +0.461 [+0.259, +0.599], positive in 100% of rung resamples.

This is a claim about composition, and composition shifts accordingly. The empty-source share of illegal moves falls from 0.230 to 0.110 while the check share rises from 0.307 to 0.448; geometry moves only from 0.285 to 0.335. The shift is substantially one class collapsing into another rather than diffuse rebalancing. Within models, check violations also concentrate at depth: their share rises from early to late game in 6 of 6 rungs, from 0.260 to 0.519 at the largest.

Nothing in this section uses model internals. The measurement is a rules engine and a move counter, which makes it reproducible against any chess model and immune to objections about

Table 1: Absolute failure rate per scored position at the smallest and largest rung, mean over seeds, and the size exponent with its rung-resampled 95% interval. All 18 runs, from the corrected pass of Section 4. Classes are grouped by what decides them: the first four are decided by the move’s own two squares, a blocked path needs one further square, and a check violation needs many. The last two rows are not disjoint classes and must not be summed with the others, the first being their total and the second a cross-cutting flag computed with a probe (Section 7).

class	3.4M	39.9M	fold drop	exponent [95% CI]
empty source	0.0583	0.0146	4.0	-0.570 [-0.756, -0.319]
opponent source	0.0092	0.0031	3.0	-0.412 [-0.595, -0.187]
own destination	0.0320	0.0106	3.0	-0.452 [-0.462, -0.442]
castling	0.0037	0.0005	7.0	-0.778 [-1.068, -0.436]
geometry, impossible	0.0590	0.0323	1.8	-0.254 [-0.313, -0.220]
geometry, blocked path	0.0127	0.0118	1.1	-0.053 [-0.228, +0.126]
leaves check	0.0779	0.0593	1.3	-0.114 [-0.153, -0.066]
all illegal moves	0.2528	0.1323	1.9	-0.267 [-0.307, -0.216]
<i>despite correct local belief</i>	0.0388	0.0404	1.0	+0.027 [-0.008, +0.097]

probe validity.

Why we do not extrapolate the fitted lines. The obvious next step is to extend the fitted lines beyond our largest model and report what they imply at, say, one billion parameters. We did this in an earlier version of this analysis and it is wrong, for a reason that applies to any per-category scaling fit and that we have not seen stated.

The categories partition the illegal moves. Their absolute rates must therefore sum to the illegal-move rate at every scale. Five independently fitted power laws with five different exponents cannot satisfy that identity away from the fitted range: the shallowest exponent must eventually dominate a total that is itself falling faster. The failure is not gradual. Summing the independently fitted parts and dividing by the independently fitted whole gives 1.02 at our largest rung, which is fit noise, but 1.27 at one billion parameters [1.13, 1.57, above one in 100% of resamples] and 2.82 at one trillion. Parts holding nearly three times the whole is not a wide estimate, it is an arithmetic impossibility, and it means the extrapolation was never reporting the exponents’ implications at all. It is worth noting that the violation runs in the direction that flatters our argument, since the categories we claim survive are exactly the shallow ones that grow without bound.

A closed compositional fit. The repair is to fit the composition on the simplex rather than one category at a time. The simplest repair is to stop fitting the aggregate and define it as the sum of the categories, which satisfies the identity at every scale and keeps every category exponent. Dividing through, the implied shares are a softmax linear in $\log N$, so the compositional form and the derived total are the same model and not a further choice. What the repair costs is the aggregate exponent: the derived total is not a power law, and its effective slope drifts with scale. Coherence is not bought with fit quality. In range the closed model attains RMSE 0.01251 across the 18 runs against 0.01322 for the uncoupled fits, so it is at least as good on the data while being the only one of the two that can be extended.

Under the closed model the check category’s share rises from 0.444 at our largest rung to 0.588 at one billion and 0.769 at one trillion, increasing monotonically toward its upper bound of one

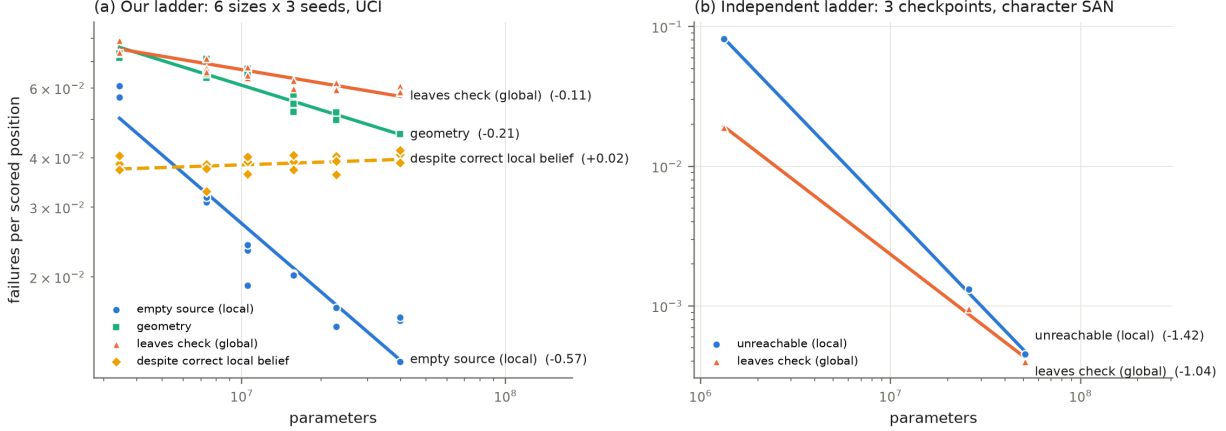


Figure 1: Absolute failure rate against parameter count, log-log, so each line is a power law and its slope, shown in each label, is the exponent. (a) Our ladder; each point is one run. (b) An independent character-level ladder (Section 6); each point is one checkpoint. In both, the local class falls faster than the global one.

as the shallowest exponent comes to dominate, while the locally decidable categories fall to 0.001. The paper’s qualitative claim is unchanged. What changes is that the projection is now bounded by construction, so the figures above are statements the categories can jointly satisfy, which the earlier ones were not.

The same discipline applies to the policy share, the fraction of illegal moves occurring despite correctly decoded local belief. It is not a member of the partition, since it cuts across the classes, but it is a proportion, and fitting a proportion linearly in $\log N$ is unbounded in exactly the same way. Fitted on the logit it rises from 0.308 at our largest rung to 0.596 at one billion and 0.951 at one trillion, having been observed between 0.148 and 0.321. We report it separately rather than folding it into the simplex; an earlier version of this analysis did fold it in, which is wrong and which manufactured a spurious interior maximum in the check category.

The failure is forced, not particular to us. Let the parts have exponents a_i and the total exponent b , and let $a^* = \max_i a_i$. The sum of the parts is asymptotically N^{a^*} and the total is N^b , so the ratio grows like N^{a^*-b} and diverges precisely when $a^* > b$. This is not a coincidence to be checked case by case. Fitted in range, b tracks the rate-weighted mean of the a_i , and a maximum is at least a weighted mean, with equality only if every exponent is identical. Our own values make the point: a^* is -0.112 , the weighted mean is -0.266 , and the separately fitted b is -0.267 , giving divergence $+0.154$.

The claim this supports needs stating carefully, because a stronger version does not follow. Divergence requires $a^* > b$, and nothing forces that for an arbitrary b ; a total assigned a sufficiently shallow exponent by hand would not diverge. What makes the inconsistency generic is how b is obtained in practice. An aggregate fitted over the same range as the parts tracks their rate-weighted mean, which lies at or below the maximum and equals it only when every exponent is identical, so $a^* > b$ holds whenever the exponents genuinely differ. The underlying mathematical point is simpler and does not depend on the fitting procedure at all: a sum of positive power laws with differing exponents does not itself have a constant power-law exponent, so an exact total and a single fitted exponent for it are different objects. Reporting both is what creates the contradiction,

and the resolution is to derive the total rather than fit it. For our own numbers the scale at which the two descriptions part company is computable from the reported exponents and coefficients together, since where the curves cross depends on their intercepts as well as their slopes: ours passes a ten percent overspend at 174M parameters, well below the billion the earlier projection used.

We checked this on a model family we did not train. Refitting Karvonen’s ladder over its own 4 classes gives divergence +0.234, an overspend of 1.28 at one billion parameters, and a breakdown scale of 243M. A different tokenisation, a different taxonomy, a different training set and a different laboratory reproduce the defect at the same order of magnitude, which is what the argument above predicts and what makes us report it as a property of the method rather than of our data.

6 External Validity

An independent ladder. Karvonen’s lichess checkpoints (Karvonen, 2024) differ from ours in every respect we could change without training: character-level PGN rather than one token per move, standard algebraic notation rather than UCI, roughly 16M training games rather than ours, and different training code. We use its 1.3M, 25.8M, and 51.0M checkpoints and read the model’s move by greedy character decoding at move boundaries.

Verifying the adapters. Every step between an external model and our measurements fails silently rather than with an error: a wrong vocabulary, a misaligned board, or a missed weight transpose all yield plausible output. Each was therefore checked before use and the checks rerun and recorded. The released weights ship without a tokenizer, so the 32-character vocabulary was reconstructed and accepted only after the model continued 3 of 3 standard openings legally. Board state was aligned to the character before each move and verified three ways at 4,081 sites, by independent replay, by re-parsing the text prefix, and by the boundary character, with 0 disagreements. Checkpoints stored in a different weight layout were converted, and all three trained checkpoints then played 20 of 20 moves legally from the opening, while the randomly initialised control failed at its first move, as it should. A cached decoder used for speed matched the uncached one at all 364 sites tested, the algebraic-notation classifier passed 8 of 8 hand-written cases, and our OthelloGPT forward pass chose a legal move at 672 of 672 positions.

Algebraic notation changes the taxonomy in an instructive way. It never names a source square, so the empty-source class cannot occur: a model that does not name the square cannot place a piece on an empty one. The nearest local class is *unreachable*, no piece of the named type can reach the destination, which also absorbs path blocking. That makes the local class less local than ours and biases the comparison against the result.

The ordering holds (Table 2, Figure 1b). Unreachable errors fall with exponent -1.415 and check violations with -1.044 , a differential of $+0.371$, against $+0.461$ on our ladder. The local share of failures falls from 0.654 to 0.377 while the check share rises from 0.152 to 0.334. Within the smallest checkpoint the check share rises from 0.060 to 0.266 across depth while the unreachable share falls from 0.730 to 0.553. At comparable size this ladder fails far less often than ours, because it was trained on far more data, so only exponents are comparable and never levels.

A game with non-local rules. Othello legality is non-local by construction: a move is legal only if it flanks a line of opponent discs, so it is never decidable from the target square. There is no published Othello ladder, so no exponent can be fitted, and we test only what the claim implies for a single well-trained model: that its residual failures should be the non-local kind. In

Table 2: Independent ladder. Absolute rates per scored position. With three checkpoints the uncertainty in the fit itself is poorly determined, and the intervals propagate only each checkpoint’s sampling noise in class shares, so they are a lower bound on uncertainty rather than a confidence statement about the slope. Deleting each checkpoint in turn leaves three two-point fits whose differential spans $[+0.303, +0.381]$, same sign throughout; we report that spread in place of an interval we cannot honestly construct.

checkpoint	illegal	failures	unreachable	leaves check
1.3M	0.1253	2,836	0.08193	0.01901
25.8M	0.0028	233	0.00132	0.00095
51.0M	0.0012	101	0.00045	0.00040
exponent	$-1.415 [-1.482, -1.365]$		$-1.044 [-1.121, -0.978]$	

Table 3: The three mechanism measurements on a model trained on roughly 16M games. Belief consistency is the fraction of illegal moves legal on the probe-decoded board. The AUC on our models is a mean over depth buckets and on Chess-GPT is pooled over depth, which is not the same quantity; only the sign and rough size of the move-touched advantage are compared. Chess-GPT’s move-touched square is the destination, since algebraic notation names no source.

	ours	Chess-GPT, human	Chess-GPT, random play
belief consistency, own board	0.355	0.122	0.188
mismatched board	0.050	0.038	0.061
random move	0.007	0.000	0.003
own / mismatched	7.2	3.2	3.1
AUC, whole board	0.587	0.530	0.485
AUC, move-touched	0.647	0.608	0.552
advantage	+0.060	+0.078	+0.068
edit, state direction	n/a	−0.078	−0.042
edit, norm-matched random		−0.005	−0.005
edit, manipulation check		1.000	1.000

Li’s synthetic OthelloGPT (Li et al., 2023), across 211,815 positions and 161 failures, 0 are moves onto an occupied square and 160 are moves that flank nothing. The remaining 1 is a pass token emitted where passing was not available, which is neither kind; we count it rather than round it away, which is why the non-local share is 160 of 161 and not all of them. The comparison is weaker than it looks, because discs never leave a square: whether a square is occupied reduces to whether it appears in the move history, which requires no state tracking at all. Failures also concentrate late, 6, 32, and 123 across the three move bands.

Heavy training. The mechanism in Section 7 rests on three measurements taken on models trained on about a hundred thousand games. We repeated all three on Karvonen’s character-level model trained on roughly 16M games (Table 3). Its board state at depth is better than ours, so the comparison is not against a model whose state was hard to extract: probe accuracy on occupied squares is 0.731 early and 0.315 late, against 0.777 and 0.167 for ours. Human games give only 131 illegal moves, so we also report uniformly random legal play, which lifts the illegal rate from 0.0026 to 0.0231.

All three hold, at reduced size. The model’s illegal moves are legal on its own decoded board

far more often than on another position’s, at a ratio of 3.1 against 7.2 for ours. Fidelity at the square the move targets predicts illegality where whole-board fidelity is near chance. And emptying a square in the model’s belief lowers the probability of moves from it, by -0.037 $[-0.050, -0.026]$ beyond a norm-matched random edit, with the decoded square flipped in 1.000 of cases. Heavy training weakens coherent error without removing it.

7 Mechanism

Why does scale leave the global class behind? Three measurements, using the probe and edit instruments of Rathore et al. (2026), point to the same answer.

Coupling is local. We repair the model’s belief at the squares its illegal move touches by adding probe-derived occupancy directions to the residual stream, and score the change in a forced-choice preference $r = P(m^*)/(P(m^*) + P(m_{\text{ill}}))$ between the best legal move and the illegal one. The ratio cancels the softmax normalising constant, so it is unaffected by a uniform rescaling of all probabilities, but it is not invariant to temperature: $r_T = \sigma((z_{m^*} - z_{m_{\text{ill}}})/T)$ moves toward one half as T rises. We state this because an earlier version claimed invariance to flattening in general, which is wrong, and because temperature alone cannot explain the disagreement between metrics reported below: rescaling logits does not reorder them, so it cannot change which move is top-1. An edit that produces more legal top-1 moves must be reordering logits, not merely flattening them, and distinguishing the two needs a temperature-only control that we have not run of the distribution. Two controls matter. A wrong-target edit installs the same occupancy at an unrelated square; an irrelevant edit spends an identical budget on squares that are wrongly decoded but not touched by the move. Both are matched to correct repair in magnitude and construction, and all contrasts are taken on positions where the two edits changed the output entropy comparably. Across 18 models, correct repair beats wrong-target in 18 (sign test 7.6×10^{-6}) and beats the irrelevant edit in 18 (7.6×10^{-6}), mean advantage $+0.039$. Only 11 of the irrelevant-edit contrasts exclude zero individually, because matching retains about half the positions per run; the evidence is that none of 18 is negative. Correcting the squares the move depends on changes behaviour more than correcting other wrong squares, which reconciles a weak whole-board coupling (Balogh and Jelasity, 2026) with a strong local one.

Correction reaches about two squares. The repair channel works: when we corrupt a single square ourselves and repair it, the model recovers a legal move in 0.733 $[0.699, 0.770]$ of cases against 0.253 for a random repair. On the model’s own failures, repair at the touched squares restores a legal move in only 0.114, a 6.4-fold gap. Natural failures involve a median of 14 wrongly decoded squares, so we tried repairing all of them. It fails in a specific way (Table 4): a two-square edit restores the decoded value at 0.677 of its targets, while an all-square edit restores 0.304, and full strength per square does no better at 0.279. Whole-board repair lowers occupied-square decoding accuracy from 0.326 to 0.185. The instrument succeeds more reliably on small corrections and does not install a reliable whole-board correction at any magnitude we tested. We tested two sizes, two squares and all divergent squares, so this bounds the procedure at those two points and is not a capacity curve; establishing one would need intermediate edit sizes at matched target difficulty and a stated rule for choosing edit strength.

What local correction cannot reach. Where decoded belief is already correct at both squares a move touches, a check violation accounts for 0.477 of failures on average against 0.286 for geometry,

Table 4: Mechanism summary. Meta rows count models out of 18; repair and fidelity rows are from a single model.

correct beats wrong-target edit, models	18 / 18
correct beats irrelevant-square edit, models	18 / 18
induced single-square corruption, recovered	0.733
random repair	0.253
natural failures, recovered by touched-square repair	0.114
decoded value restored, two-square edit	0.677
decoded value restored, all-square edit	0.304
occupied-square accuracy, before and after all-square edit	0.326 to 0.185

and is the largest class in 15 of 18 models. Such failures cannot be attributed to the decoded belief being wrong at the move’s squares. Their absolute rate is flat with size, exponent +0.023 [−0.005, +0.066], a differential against empty-source errors of +0.596 [+0.367, +0.761].

The flat rate is two effects cancelling, not one effect absent. An earlier version of this paper read that flat exponent as the underlying failure propensity being untouched by scale. It does not support that, and the quantity is a joint event:

$$P(\text{illegalcorrect local decoding}) = P(\text{correct local decoding}) \cdot P(\text{illegal} \mid \text{correct local decoding}).$$

Both factors move. Across the 18 runs the marginal rises with exponent +0.148 [+0.115, +0.182], decoding accuracy going from 0.471 to 0.675, and the conditional falls with exponent −0.125 [−0.182, −0.053], whose interval excludes zero. The joint rate is flat because the two nearly cancel. The decomposition is exact rather than estimated, so this is a restatement of the same number and not a competing measurement.

The correct reading is therefore the opposite of the one we drew: scale does reduce the rate at which a model fails while holding correct local state, and the flat joint rate conceals it because a better model decodes correctly more often and so has more occasions on which to fail that way. What survives is the weaker claim that this component declines more slowly than the locally decidable classes, which is what the differential measures. An earlier oracle-free attempt to repair state by re-presenting information the model already attends to also failed, leaving the late-game illegal rate at 0.295 against 0.296 without intervention.

We do not claim these failures are unrepairable in principle. The instrument cannot install a multi-square correction, so whether the residual is genuine policy failure or simply beyond its reach is undetermined, and we recorded the stronger reading as retracted (Appendix B). The claim is narrower and sufficient: the component scale leaves behind is concentrated in exactly the failures that correction at the move’s squares cannot address.

8 The Standard Metric Inverts

Interventions on state are usually scored by whether the model’s output becomes legal. On our repair edits that metric orders the conditions nearly in reverse (Table 5). On one model, correct repair produces the largest preference shift, +0.140, and a random direction the smallest, +0.041; yet after the edit the random direction leaves a legal top-1 move in 0.268 of cases and correct

Table 5: Repair conditions on one model (600 positions, 61 games): change in forced-choice preference, fraction of top-1 moves legal after the edit, change in entropy, and how often the target square’s decoded value flipped. The null edit re-installs values already decoded correctly and is zero by construction.

condition	Δr	legal after edit	Δ entropy	decode flipped
correct repair	+0.140	0.147	−0.017	0.826
irrelevant squares	+0.061	0.203	−0.177	0.163
wrong target	+0.053	0.037	−0.007	0.171
random direction	+0.041	0.268	+0.298	0.080
null	0.000	0.000	0.000	0.000

repair in 0.147. The random direction also raises output entropy most, by +0.298. Hedging moves probability off a confidently illegal move and onto whatever else is plausible, some of which is legal.

This is not one model. In 18 of 18 models the random direction produces more legal moves than correct repair while correct repair produces the larger preference shift (sign test 7.6×10^{-6}), and the random direction has the best legal-move rate of all four edits in 16 of them. Its advantage in legal moves over correct repair ranges from +0.048 to +0.143.

The same artefact affects our own preference metric if uncontrolled. The raw advantage of correct repair over wrong-target is +0.087; restricted to positions where both edits changed entropy comparably it is +0.036 [+0.011, +0.058], 2.4 times smaller. The edit also responds gradedly to strength: preference rises from +0.042 at a quarter of calibrated strength to +0.140 at calibrated strength, then turns to −0.046 and −0.091 at double and quadruple, a rise-then-collapse that is harder to explain as generic damage than a monotone decline would be. Studies that score state interventions by illegal-move rate, or by any error rate a flatter distribution can lower, need not be measuring what it is taken to measure. We do not claim the mechanism is distributional flattening: temperature alone cannot reorder logits and so cannot change which move is top-1, and the random edits do change it, so something other than flattening is at work and we have not identified it.

9 Discussion

The practical content is about where effort goes. The class that does not fall with size is the one defined by constraints spanning the whole board, and on our evidence it is also the class that correcting the model’s belief about a move’s own squares does not reach. We read this, as an interpretation we have not tested directly, as a caution for interventions such as auxiliary board losses and state supervision: they aim at the model’s knowledge of where pieces stand, and the failures most directly attributable to that knowledge are the ones scale is already removing. An intervention that lowers total error while leaving the global class flat may have improved only what scale improves anyway, and should be evaluated by its effect on that class.

This also bounds what an intervention study can show. A training comparison that targets state-attributable failure is adequately powered on our data, detecting a change of 19% of that component with seed variation contributing little (relative spread 0.028), but the component is a small share of what goes wrong, so a detectable improvement is a small behavioural one.

A methodological point generalises further than our subject. Whenever failures are decomposed into categories and each category is given its own scaling exponent, the fits sit alongside a separately fitted aggregate, the two make claims that cannot both hold, and extrapolation is where the conflict becomes visible. The check costs nothing: sum the extrapolated parts and see whether they still

make the whole. We report it because our own uncoupled fits passed every gate in this paper, agreed with the data in range, and were still projecting an impossibility two orders of magnitude out. Any decomposition of error into categories that scale at different rates, whether by capability, benchmark subset, or error type, has this property, and fitting on the simplex removes it without costing accuracy. We supply the check as `code/closure_check.py`, which needs only the exponents and intercepts such papers already print.

One consequence is easy to miss, and we missed it. An inadmissible fit voids negative conclusions as well as positive ones. While preparing this section we tested whether failure composition is governed by achieved competence rather than by size, rejected it because the fitted shares exceeded one, and recorded it as refuted. That rejection was worthless: predicting an impossibility is what the uncoupled form does everywhere, so its disagreement with data carries no evidence about the hypothesis under test. Refitting the same comparison on the simplex, in competence coordinates, and predicting Karvonen’s checkpoints out of sample does refute it, by a total variation distance of 0.490 against 0.100 for size coordinates and 0.115 for predicting our own mean composition. The conclusion survived; the first argument for it did not. A model that cannot make an admissible prediction cannot be used to reject anything.

The local and global distinction is not specific to chess. Wherever a checker can say not only that a step is wrong but which constraint it violated, program execution, spreadsheet recalculation, and formal proof among them, the same decomposition is available, and the question of which violations scale removes can be asked without any access to model internals.

10 Implications for Individuals, Organisations, and Society

For the people on the receiving end of a long-horizon failure. The errors this paper separates are not equally visible to the person affected. An error decidable from the move’s own squares is the kind a user can usually catch: it contradicts something in front of them. An error that is locally consistent and globally wrong is not, because every part of it looks right. Our measurements say scale removes the first kind several times faster than the second, so the errors that survive in larger systems are disproportionately the ones a non-expert cannot see. That shifts the burden of catching mistakes from the person who can check a detail to the person who can check the whole, and in most deployments those are not the same person and the second is not present.

For managers and organisations. The practical reading is about where a mitigation budget goes. A class of failure that scale removes on its own is a poor target for engineering effort, and a class with an exponent near zero will still be there after the next model upgrade. Our estimate is that errors surviving a correct local belief have a flat joint rate, exponent $+0.023$, which decomposes into a rising chance of decoding the position correctly and a conditional failure rate falling with exponent -0.125 . The improvement in the conditional rate is real but is cancelled in the headline number, so an organisation reading only the joint rate will conclude wrongly that scale does nothing for this class. We do not read the conditional decline as a measurement of planning ability: correct decoding at two squares is not a correct global state, and the group being conditioned on changes composition across models, so this is a conditional association rather than an isolated capability. An organisation deciding between waiting for the next generation and funding a mitigation can, in a domain with a checker, measure which class its own failures fall into rather than arguing about it. The second lesson is about measurement: illegal-move rate, the metric most readily available, ranks a norm-matched random edit above a correct repair in 18 of 18 models, while the repair produces

the larger shift in a fixed forced choice. Two reasonable measures of the same intervention disagree, so a single headline number cannot settle whether an intervention helped.

Strategic implications. Treat the composition of failure, rather than its rate, as the quantity worth tracking. A falling error rate is compatible with the dangerous share of errors rising, and our ladder shows exactly that: the total falls while the fraction that is globally rather than locally wrong increases. An organisation that reports only the aggregate will see improvement in the period when its remaining failures are becoming harder to detect. The capability to decompose is cheap where an oracle exists and needs no access to model internals, so it is available to a buyer as well as to a vendor, which matters when the two disagree about whether a known defect will be fixed by scale.

For society and the public record. The decomposition used here needs a checker that can say not only that an output is wrong but which constraint it violated. Chess has one for free. Program execution, spreadsheet recalculation and formal proof have one too, and in those domains the same question can be asked by anyone, at no cost, without the cooperation of whoever built the system. Where no such checker exists, and most consequential deployments are in that position, claims about which errors scale away cannot currently be audited by anybody. We would rather state that plainly than let a result measured in a game stand in for domains that cannot yet be measured.

11 Limitations

Depth and width are confounded. Both ladders grow depth and width together, so an exponent in parameters does not isolate either.

Three external checkpoints. The independent ladder supports directional agreement, not a second estimate of the slope; its intervals omit fit uncertainty.

Notation changes the classes. Algebraic notation removes the empty-source class, and its nearest local analogue absorbs path blocking.

Othello cannot test scaling. No ladder exists, and its local class requires no state tracking, so it is a consistency check rather than a replication.

Consistency is not validation. The derived total guarantees the projection is arithmetically possible, which the uncoupled fits are not. It does not make it observed: every scale past 39.9M is extrapolation, and a different link satisfying the same constraint would give different numbers while agreeing that the uncoupled ones are inadmissible.

The check class is not an exact power law. Fitting it quadratically in $\log N$ gives a curvature term whose bootstrap interval excludes zero, so the exponent we report for it is a local slope over our range. This does not change its sign, its separation from the local classes, or the ordering that is this paper’s result, all of which are about the fitted range. It does mean a reader should not extend that one exponent far beyond the ladder.

The residual is unattributed. The repair instrument corrects about two squares, so whether failures despite correct local belief reflect policy or unreachable state is undetermined.

Probe-based rows depend on a linear probe. A nonlinear decoder of four times the capacity gains only +0.011 to +0.033 across depth (Appendix A), so decoder weakness does not explain decay, but the probe-based rows inherit its approximation. The headline result uses no probe.

One game family. Chess and Othello both carry exact oracles, which is what makes them measurable and what makes them unrepresentative of open-ended reasoning.

What the intervals cover, and the unit of variation. Intervals on the main ladder resample rungs, because seeds within a rung share an architecture and are not independent draws of it. This makes the rung the unit of variation and leaves within-run sampling noise over games and positions outside the interval rather than inside it. We take rung-level variation to be the larger term, but we have not decomposed the two, and a reader who wants a game-clustered interval on a single run will not find one here. The external ladder differs: its per-checkpoint class shares are bootstrapped clustering on games, so game-level noise is propagated there and the fit uncertainty is what is missing instead.

The sign tests over the 18 runs treat runs as exchangeable. Seeds within a rung satisfy that by construction, differing only in initialisation and batch order. Rungs do not: they differ systematically in size, which is the very thing under study. A sign test pooled across rungs does not establish that an effect holds at every size merely because the runs span sizes, and we separate the two statements. The observed count is a description of these runs: the effect has the same sign in all of them. The inferential claim a pooled sign test supports is the weaker one that the median effect over this set of runs is nonzero. Neither licenses a statement about a model drawn from some population of architectures, which 6 sizes could not support.

Unequal scored sets in the external table. The external checkpoints were not all scored on the same number of positions. The two larger ones share an identical set of 84,557 positions, while the smallest was scored on 22,634. The sets are nested rather than disjoint, the smaller being a prefix of the same held-out corpus, so this is a difference in sample size and not in sampling population. It still means the three rows of that table rest on unequal precision, and the smallest checkpoint is also the one with by far the most failures, so its shares are the best determined despite the smaller set. We also note that the game count reported for each checkpoint is the number of games contributing at least one failure, not the number scored, which is why it falls as the models improve.

12 Conclusion

Scale does not remove long-horizon errors evenly. In chess transformers it removes errors decidable from a move’s own squares several times faster than errors decided by the whole board, the ordering replicates on an independent model family, and the failures that occur despite correctly decoded belief at the move’s squares do not decline at all. The measurement needs only a rules engine. What it suggests is that the failures worth studying at scale are the ones defined by global constraints, and that interventions should be judged by their effect on those, measured with a metric a flatter output distribution cannot improve.

Declarations

Disclosure statement. The authors report no conflict of interest.

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability statement. The analysis code, every result file cited by the manuscript, and the macro provenance record are available at <https://github.com/MohitPratapSinghRathore/>

derived-total. Trained checkpoints are available from the corresponding author on reasonable request, being too large to distribute with the repository. External checkpoints are public.

AI usage declaration. During the preparation of this manuscript the authors used a generative AI tool to assist with language editing and refinement of the text, and with writing analysis code. The authors reviewed and edited all output and take full responsibility for the content of the paper. Generative AI was not used to generate or analyse the study’s data, which were produced entirely by the measurement pipeline described in the paper.

A Probe capacity

Every fidelity curve in this line of work is read with a linear probe, which could confound information leaving the representation with information becoming nonlinearly encoded. On identical activations and budgets, a two-layer probe whose hidden layer is four times the residual width improves occupied-square accuracy by between +0.011 and +0.033 across depth, with no tendency to widen. The same nonlinear probe reaches 0.000 on a control task with shuffled boards and 0.015 on a randomly initialised model in the deepest bucket, so there is no memorisation headroom for extra capacity to exploit. Decay in these curves is therefore not explained by the weakness of a linear decoder, though a single alternative decoder cannot rule out every nonlinear encoding.

B Integrity record

The analysis plan was frozen and hashed (SHA-256 prefix `28b68e5200dc0dba`). We state plainly what that does and does not establish, because an earlier version of this statement claimed more than the record supports. A hash fixes a document’s content; it carries no time, so on its own it cannot show that the plan preceded the analyses it governs. The ordering evidence is the commit history of the public repository, which is self-maintained and not a third-party trusted timestamp, and we do not ask the reader to treat it as one.

What that history shows is a staged design rather than a single registration before all analysis. The plan was frozen on 2026-09-05. The failure-classification results on which the central claim rests first enter the repository on 2026-09-06, after the freeze. One set of results does not: the week-zero go/no-go measurements that decided which of the candidate studies was viable were committed on 2026-09-05, shortly before it. Those were scoping measurements and the plan was written in light of them, so the freeze pre-registers the main programme and not the scoping phase. We would rather say this than describe the whole project as pre-registered and leave a reader to discover the four minutes for themselves. Three amendments changed reported claims and are stated here in full. First, the frozen entropy-matching procedure was vacuous, since weighting strata by their own counts reproduces the overall mean; it was replaced by matching on positions after point estimates were seen, and the replacement reduced the headline effect. Second, a prior that about a third of illegal moves would be restored to legality by repair was mis-specified, by conflating whether an error is explicable by wrong belief with whether repairing belief is sufficient to change behaviour; the observed rate was far lower. Third, a conclusion that the failures repair could not fix are genuine was retracted once whole-board repair was shown not to install a board, which is why Section 7 leaves that residual unattributed.

Two further corrections changed how a result was stated rather than a reported value. The scaling result was first framed in terms of shares of failures, and that framing was stated before it was found to imply that models grew worse at a class they were in fact improving at; it was revised

to absolute rates, which is the form reported here. The metric comparison was first reported on one model and is now reported across all of them. Analysis corrections that changed no reported claim, including a move-touched definition that scored illegal algebraic moves at arbitrary squares and edit intervals clustered on too few games, are recorded in the analysis repository’s version history.

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